

PrimeSim™ XA Help: What's New

Version W-2024.09, September 2024



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What's New in This Release

PrimeSim XA version W-2024.09 includes the following new features and enhancements:

- [Enhancement to the HSPICE Device Models](#)
- [Enhancement to the Spectre Simulator Device Models](#)
- [Enhancement to the Eldo Simulator Device Models](#)
- [PrimeSim XA - New, Changed and Obsolete Commands and Options](#)
- [PrimeSim XA - Additional Product and Documentation Enhancements](#)
- [VCS® PrimeSim™ AMS Features and Enhancements](#)

Enhancement to the HSPICE Device Models

In version W-2024.09, the PrimeSim XA tool supports the following new device models:

- MOSFET Models:
 - Level 70, BSIMSOI4 Model, version 100.1
 - Level 73, HiSIM_HV Model, version 2.5.1

For more information, see the Device Models section in *PrimeSim XA User Guide*.

Enhancement to the Spectre Simulator Device Models

In version W-2024.09, the PrimeSim XA tool supports the following new spectre simulator device models:

- MOSFET Device Models: BSIMSOI100 Model

For more information, see the Supported Spectre Simulator Device Models section in *PrimeSim XA User Guide*.

Enhancement to the Eldo Simulator Device Models

In version W-2024.09, the PrimeSim XA tool supports the following new eldo simulator device models:

- Eldo MOSFET Models: Level 72, BSIMSOI4.0 to 4.3 Model
- Eldo MOSFET Models: Level 79, BSIMSOI4.4 to 4.6 Model, BSIMSOI100 Model

For more information, see the Supported Eldo Simulator Device Models section in *PrimeSim XA User Guide*.

PrimeSim XA - New, Changed and Obsolete Commands and Options

The following table lists the commands and command-line options that are new or changed in this release.

Table 1 *PrimeSim XA - New and Changed Commands in This Release*

Command	Description
<code>.dd_check_block_power</code>	Checks the block power.
<code>iset_speed_scale</code>	Dynamically scales the transient simulation engine tolerance in interactive mode.
<code>cck_analog_pdown</code>	Added new <code>-idsth_hizdev1 <value></code> argument that specifies a separate current threshold for paths that contain HiZ-gated MOSFETs.

For more information, see the *PrimeSim™ XA User Guide* and *PrimeSim™ XA Command Reference*.

PrimeSim XA - Additional Product and Documentation Enhancements

- Added the `vcd2vec` command information in the *PrimeSim™ XA Command Reference* manual.

VCS® PrimeSim™ AMS Features and Enhancements

In version W-2024.09, the VCS PrimeSim AMS tool provides the following enhancements:

- [VCS PrimeSim AMS - New, Changed and Obsolete Commands and Options](#)
- [VCS PrimeSim AMS - Additional Product and Documentation Enhancements](#)

VCS PrimeSim AMS - New, Changed and Obsolete Commands and Options

The following table lists the commands and command-line options that are new or changed in this release.

Table 2 *New and Changed Commands in This Release*

Command	Description
<code>udn_bidir</code>	Controls the behavior of an <code>udn_bidir</code> interface element. This command sets a Verilog-A module for a specified user-defined nettype on a specific node.
<code>udn_e2n</code>	Added new arguments <code>-min_delta value</code> and <code>-max_delta value</code> .
<code>udn_n2e</code>	Added new argument <code>-min_delta value</code> .

For more information, see the *VCS® PrimeSim™ AMS User Guide*.

VCS PrimeSim AMS - Additional Product and Documentation Enhancements

- Added the following new topics in the *VCS® PrimeSim™ AMS User Guide*:
 - Introducing the VCS PrimeSim AMS Tool
 - Running Cosimulation
 - Pausing and Resuming Simulations
 - Debugging Current Consumption

Following are the new sections that are added to the “User-Defined Nettypes With Custom Interface Elements” topic of *VCS® PrimeSim™ AMS User Guide*.

Chapter 1: What's New in This Release

VCS® PrimeSim™ AMS Features and Enhancements

- Placement of the Verilog-A Interface Elements
- Resolution Function
- Waveform Names
- Example of Unidirectional and Bidirectional Interface Elements
- Commands to Change the Arguments of the Interface Elements